Case Study Rubric

Due: see Canvas

Submission format: One PDF report and one code file submitted to Canvas

Individual Assignment

General Description: Submit a single PDF report and your code file to Canvas.

Why am I doing this? This case study gives you the opportunity to apply core data science skills – including model building, feature engineering, and error analysis – to a real-world infrastructure problem with genuine public stakes. As you work through this assignment, you will see how data-driven analysis can be applied to complex, interdisciplinary challenges at the intersection of energy, weather, and public policy.

What am I going to do? By this point in the course, you have developed a working knowledge of data cleaning, exploratory analysis, and predictive modeling. This case study asks you to bring those skills together into a single, independently-driven project delivered as one PDF report. You will begin by loading and exploring hourly electricity data alongside daily weather observations, identifying patterns, anomalies, and relationships between the two datasets. From there, you will engineer any features you deem relevant. You will then build two demand-forecasting time-series models: a baseline model using only past demand data, and a weather-augmented model using past demand data and corresponding weather data. Finally, you will interpret your results and draw an evidence-based conclusion about whether data meaningfully improves forecast accuracy, and what those findings imply for grid operators facing real capacity decisions.

How will I know that I have succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Specification	**Details**

Formatting	• Code file - one code file submitted to Canvas (.py, .R, or .ipynb)
	• Written report - one PDF reported submitted to Canvas
	    o Written in 12 pt Times New Roman (or comparable font), with standard margins and double spacing
	    o About 4-6 pages in length, inclusive of figures and tables

Code File	• <u>Goal</u>: Your code file should be a complete, self-contained script that reproduces all results, figures, and tables presented in your report. • Submit as a .py or .R file (or, alternatively, a .ipynb notebook) • Include a header comment at the top identifying the author, course, and a brief descriptions of what the script does. • **Include copious inline comments** explaining what each step does and why. A reader unfamiliar with your project should be able to follow your logic throughout.

<table>
<tr>
<td>Report</td>
<td>

<u>Goal:</u> This report is your complete deliverable, integrating your exploratory analysis, visualization, modeling, and recommendation into one document.
The reader should understand the problem and your solution in the context of that problem.
The model to implement is left up to you, but you must select a model which can accommodate weather variables alongside past time-series demand data.
The report should include the following sections:

Introduction (1-2 paragraphs): use AT LEAST the provided sources to establish problem context and motivate the analysis.
Exploratory Data Analysis (1-2 paragraphs + figures): load and describe the electricity and weather datasets; include at least three figures examining distributions and/or relationships of key variables (e.g., demand over time, temperature vs. demand, seasonal patterns)
Methods (1-2 paragraphs): briefly explain the design of both the baseline and weather-augmented models, explaining any key preprocessing and feature engineering steps.
Results (1 paragraph + figure + table): evaluate and compare the models using appropriate error metrics reported in a table; include at least one figure showing predicted vs. actual demand for each model.
Conclusion (1-2 paragraphs): offer a clear, evidence-based recommendation on whether weather data improves forecast accuracy and what the implications are for grid operators.

Claims about the model or data must have corresponding references to figures or tables included in the report.
All references (using IEEE Documentation style) should be listed at the end of the document in an appropriately titled section.

</td>
</tr>
</table>

Acknowledgements: Special thanks to Professor Karsten Siller for providing the template of this rubric document.